CSIT883 System Analysis and Project Management - Group Project Proposal

SmartCampus: An Intelligent Campus System for GlobalConnect University

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1. Project Introduction

GlobalConnect University (GCU) serves over 30,000 full-time and 8,000 part-time students with 2,500 faculty and staff. Its new strategic plan targets better learning outcomes, lower administrative cost, and fuller use of Web-based and AI-based technology. Today, however, learning support, facility management and administrative services run as isolated systems: students cannot get timely guidance, classrooms and laboratories are booked inefficiently and consume energy when unoccupied, and document-related administration still relies on manual handling.

SmartCampus is an integrated intelligent campus system that unifies these three service domains behind one conversational, knowledge-driven interface. It comprises **three subsystems and six modules**, four of which embed AI techniques, and it is built to be demonstrated live in class.

2. Our Innovative Idea

One shared, grounded LLM core instead of isolated smart tools. The project's central innovation is a single LLM service layer - a retrieval-augmented generation (RAG) core plus an intent-routing gateway - that all three subsystems call through the same API. A student asks one chatbot about a course, then about a free laboratory, then about a transcript, without changing system. This turns three separate departmental tools into one intelligent campus.

Grounded, verifiable answers. The LLM never answers from memory alone: it retrieves from GCU's course catalogue, policy documents, timetable and the student's own record, and every reply cites the source it used. Answers that cannot be grounded are refused and escalated to a human service desk, so the system cannot invent academic rules.

Runs offline for the live demonstration. The core is model-agnostic: it uses a hosted LLM in normal operation and automatically falls back to a locally hosted open-source model plus a retrieval-only FAQ mode when the network or API quota is unavailable. The demonstration therefore never depends on an external service.

Privacy-preserving personalisation. Student-profile data is used only through pseudonymised feature vectors, so risk warnings and recommendations are personalised without exposing a student's identity to the module that generates them.

One AI capability, reused across subsystems. The same LLM document-understanding component that pre-checks a certificate application also classifies maintenance fault reports, so a single investment in the AI core raises service quality in all three subsystems.

3. System Functions

Subsystem 1 - Intelligent Learning Support (improves learning experience and academic success)

M1.1 Conversational Study Assistant - 24/7 LLM chatbot, with voice input, answering questions on courses, prerequisites, timetables, assessment rules and campus policy; supports multi-turn dialogue and cites the retrieved source. *(AI: LLM + RAG + intent classification; hosted model with a local open-source fallback)*

M1.2 Adaptive Learning-Path Recommender and At-Risk Early Warning - recommends electives and a study sequence from the student's completed courses, grades and interests, and flags students at risk from engagement and assessment trends with an actionable support suggestion. *(AI: hybrid content/collaborative-filtering recommendation + explainable ML risk classifier)*

Subsystem 2 - Intelligent Facility and Resource Management (convenient and efficient facilities)

M2.1 Smart Booking and Fault Reporting - one-step booking of classrooms, laboratories, meeting rooms and sports facilities, with predicted occupancy, clash detection, automatic alternative suggestions and a real-time availability dashboard; faults are reported in natural language and auto-classified and routed to the right maintenance team. *(AI: occupancy prediction + LLM fault classification)*

M2.2 Energy-Saving Control and Maintenance Tracking - schedules lighting, air-conditioning and equipment power from timetable and real occupancy data, predicts equipment failure from usage history, and tracks every maintenance ticket to closure with cost statistics. *(AI: ML predictive maintenance)*

Subsystem 3 - Intelligent Administrative Service (streamlined routine processes for students and staff)

M3.1 Intelligent Document Checking and Auto-Review - applies rule-based validation plus LLM understanding to transcript, certificate and document applications, reports missing or inconsistent fields and supports automated form filling and information verification. *(AI: rule engine + LLM document understanding)*

M3.2 Unified Service Portal, Tracking and Analytics - a single portal for transcript, certificate and document applications with status tracking, notification (email/in-app), and a staff workload visual-analytics dashboard for administrators.

Cross-cutting: unified login and role-based access control, the shared LLM/RAG service, a common database, and a web front end - all six modules run as one demonstrable system.

4. Planned Job Distribution

The work is divided into six roughly equivalent module responsibilities, **one owner per module**, each carrying analysis, design, implementation, testing and documentation for that module.

Member	Subsystem / Module owned	Main responsibilities	Role in team
Rongkai Cheng	1.1 Conversational Study Assistant	LLM/RAG core, dialogue and voice interface, knowledge-base ingestion, prompt and evaluation design	Project Manager (Phases 1-2); AI lead
Dequn Shang	1.2 Learning-Path Recommender and At-Risk Early Warning	Student-profile feature design, recommendation algorithm, risk model training and evaluation	Backend developer; data/AI analyst
Jun Yang	2.1 Smart Booking and Fault Reporting	Booking workflow and clash detection, occupancy prediction, availability dashboard, fault intake	Facility subsystem lead; integration
Lexi Shu	2.2 Energy-Saving Control and Maintenance Tracking	Occupancy/timetable-driven control logic, predictive-maintenance model, ticket lifecycle and costing	Project Manager (Phase 3); QA lead
Zheyuan Huai	3.1 Intelligent Document Checking and Auto-Review	Rule-based validation engine, LLM document understanding, auto form filling, test data design	Admin subsystem lead; document owner
Shiran Yuan	3.2 Unified Service Portal, Tracking and Analytics	Portal and notification service, application status tracking, workload analytics dashboard, UI/UX	Project Manager (Phase 4); UI lead

Shared duties (all members, roughly equal effort): system analysis and requirements consolidation; integration testing with the other members' modules; preparation of the progress and final presentations; and review of the final report. **Management rotates** - Rongkai Cheng leads the proposal and analysis phase, then Lexi Shu, then Shiran Yuan, so every member manages the team in turn and gains project management experience.

5. Team Roles and Schedule

Roles: Project Manager (rotating: Rongkai Cheng, then Lexi Shu, then Shiran Yuan), AI/Model Lead (Rongkai Cheng, Dequn Shang), Backend Developers (Dequn Shang, Jun Yang, Lexi Shu), Frontend/UI Lead (Shiran Yuan), System Analyst and Documentation Lead (Zheyuan Huai), Quality Assurance Lead (Lexi Shu), Integration Lead (Jun Yang). Every member owns exactly one module, and module ownership carries accountability for that module's chapter of the final report.

Schedule: 13 Oct 2026 proposal submitted; Phases 1-2 - requirements analysis, system design and LLM core prototype; Phases 3-4 - module implementation, integration and testing; **Week 12** progress presentation with live demonstration; **Week 13 (13 Dec 2026)** final report, presentation slides and deliverables (source code, test data and results) submitted.

Key risks and responses: *API quota or network failure during the demonstration* - mitigated by the local open-source model and retrieval-only fallback described in Section 2. *LLM giving an incorrect academic answer* - mitigated by grounding every reply in a retrieved source, refusing ungrounded answers and escalating them to a human. *Six modules integrating too late* - mitigated by freezing the shared LLM service interface at the end of the design phase and integrating continuously from Phase 3.